



Violence in the Household and Happiness in Mexico

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Abstract

This article has two purposes: first to investigate whether violence influences wellbeing; second to investigate whether the effect of violence is greater for women than for men. The results show that there is a significant impact on the wellbeing of individuals when suffering violence from someone with whom they live, with women being more severely affected than men.

1. Introduction

Recently, a series of studies has addressed the effects of being a victim of crime on different types of wellbeing. Most have focused on crime (Cornaglia, Feldman and Leigh 2014; Davies and Hinks 2010; Dustmann and Fasani 2016; Mahuteah and Zhu 2015). These studies found that victims of crime report lower levels of happiness than non-victims (Powdthavee 2005). The objective of this article is to investigate the impact of domestic violence on the wellbeing of individuals. The article also examines some elements that can prevent violence.

Domestic violence has repercussions for the economic level of families and countries (García-Moreno et al. 2005). Those who suffer domestic violence tend to be less productive at work. Likewise, the trauma caused to children can have social consequences. Furthermore, poor mental health can reduce employment rates (Fritjers, Johnston and Shields 2014). Based on the above, the damage to the mental wellbeing of victims of violence can be argued to have negative economic outcomes.

Some researchers have studied happiness using the theory of subjective wellbeing, evaluating the impact of several variables on satisfaction with life. There are numerous studies about family (Cracolici, Giambona and Cuffaro 2013; Glavin and Peters 2015; Qian and Qian 2015; Rojas 2007; Velasquez and Ramírez 2012), however, the impact of domestic violence on happiness has been little explored, except for a focus on teenagers (Callahan, Tolman and Saunders 2003). Hence this article seeks to contribute to the literature by exploring this under-researched area.

Violence against women is a serious problem worldwide. In Mexican society 'machismo' tradition supports the view that a

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man's violence towards his wife is a male right (Moral de la Rubia and Ramos Basurto 2016). For the indigenous population in Mexico, violence against women is justified if she 'does something wrong' (Krug et al. 2003, p. 103). Azaola (2012) describes the Mexican family as a violent social institution. Between 2007 and 2012, the death rate of women murdered in Mexico grew by more than 200 per cent, from 1.4 to 4.6 per cent of total deaths (SEGOB; INMUJERES México; ONU Mujeres México 2016).

This issue of physical violence within the household has attracted specialists from different fields of knowledge: economists have studied the effect of violence on women's income (Tauchen, Witte and Long 1991); lawyers have examined family violence prevention strategies (Fagan 1989); sociologists have investigated the social causes of domestic violence (Callahan, Tolman and Saunders 2003), anthropologists have focused on the efforts of indigenous women to combat violence in rural areas (González-Montes 2009) and psychologists have investigated the consequences for victims' emotional health (Anderson 2002). The present work takes a multidisciplinary approach, using the theoretical framework of subjective wellbeing studies.

Prior studies of subjective wellbeing have not reached consensus on whether gender has an effect on the happiness of individuals (Stutzer 2004). However, it has been documented that some variables affect men and women differently. For example, men engaging in household responsibilities may report lower levels of wellbeing (Qian and Qian 2015) and home ownership is more valued by women than by men (Hu 2013).

This article has two purposes. The first is to investigate whether violence influences wellbeing. The second is to investigate whether the effect of violence is greater for women than for men. We use a database containing information on assaults and threats from household members, as well as information on the level of happiness and satisfaction reported by individuals, of more than 10,000 people.

The results show that there is a significant impact on the wellbeing of individuals from

suffering domestic violence. With respect to gender differences, the study finds that women suffer violence more than men and that the impact of such violence affects women more severely. The study also finds that support from family and friends can decrease the incidence of abuse and increase subjective wellbeing. Consistent with the current literature, this study considers happiness and subjective wellbeing as synonyms (Rojas 2009; Winkelmann 2005). In this article we consider domestic violence as those acts of physical aggression and threats that a person is subjected to by someone who lives in the same house.

The remainder of the article is organised as follows. Section 2 describes the theoretical approaches to subjective wellbeing, the household and violence. Section 3 outlines the methodology and hypotheses. Section 4 presents the results and the final section discusses the findings and concludes.

2. Theoretical Approach

This section briefly describes the theoretical approaches adopted in this study, the concept of happiness, and some possible characteristics, causes and consequences of violence.

There are several studies that highlight the effect of being a victim of crime on the wellbeing of individuals. It has been identified that in the United States people are willing to spend an amount 40 times greater on security than crime actually costs (Cornaglia, Feldman and Leigh 2014), suggesting that crime also has a considerable emotional impact. In the United Kingdom Dustmann and Fasani (2016) identify that crime causes significant suffering and anguish for citizens, with a greater impact on the female population, suggesting that women are more sensitive to acts of violence. These findings support a need to develop policies that allow greater protection based on gender.

The research findings with respect to domestic violence suggest that, just as for other crime, domestic violence may also affect individual wellbeing and have a similar pattern across genders. It is important to note that violence has repercussions on labour income and on psychological and physical health.

2.1 Subjective Wellbeing

The present work relies on a widely adopted premise in the literature on subjective wellbeing; that each person is considered to be the best judge of his or her own wellbeing (Mackerron 2012). Studies of subjective wellbeing have found that: income has a moderate influence on happiness after individuals satisfy basic needs (Diener and Biswas-Diener 2002); unemployment affects happiness beyond income loss (Winkelmann and Winkelmann 1998); happier people are more likely to marry and to stay married (Stutzer and Frey 2006); the presence of people addicted to harmful substances impairs the happiness of the rest of the family (Charles-Leija, Aboites and Llamas 2016; Rojas 2015). Furthermore, it has been suggested that subjective wellbeing surveys can be a useful complement to the methods of contingent valuation and hedonic prices (Dustmann and Fasani 2016).

However, the impact of violence and aggression on subjective wellbeing is worthy of further exploration. In this article happiness, as measured subjectively by survey questions, is understood to measure an individual's feelings in relation to subjective enjoyment of life. The data reported by individuals are considered to be reliable (Beytia and Calvo 2011).

In economics the topic of the family has been approached theoretically since the 1960s (Becker 1965, 1973, 1974; Becker, Landes and Michael 1977), however most of this previous work has assumed that family members are motivated by altruism towards their fellow family members. More recently, authors have considered non-cooperative behaviours where each spouse maximises his or her utility subject to the restrictions that their partner generates (Farmer and Tiefenthaler 1997).

2.2 Causes, Characteristics and Consequences of Violence

Domestic violence can have multiple origins. It has been argued that violence and abuse can be a source of gratification for the aggressor or can be used for instrumental purposes and to control the

behaviour of others (González-Montes 2009; Tauchen, Witte and Long 1991). Some individuals may understand control and jealousy as love (Callahan, Tolman and Saunders 2003). Violence may also stem from stress, depression, low self-esteem and problems with drugs and/or alcohol (Anderson 2002).

Violence against women is long-established in society and many cultures are tolerant of the ill-treatment of women (Páez 2011). Women report violence more often than men and a high percentage of violence is inflicted inside the household (Callahan, Tolman and Saunders 2003; Borooah and Mangan 2009). Laws that facilitate divorce have been associated with reduction in female suicide rates and the number of women murdered by their partners (Stevenson and Wolfers 2006).

Among the negative effects of violence are depression, anxiety and low self-esteem (Holt and Espelage 2005). Domestic violence is associated with violence in dating, as well as in marriage (Callahan, Tolman and Saunders 2003). Studies have established that victims of violence are more likely to inflict it in the future (Anderson 2002). Therefore, domestic violence has implications for both individuals and for society—victims of violence are more likely to develop anti-social and criminal behaviour. Domestic violence against children has been shown to be linked to stress, anxiety, depression and delinquency (Callahan, Tolman and Saunders 2003). Other consequences for victims are physical and mental health problems; at the extreme end of the spectrum of domestic violence consequences is suicide (Anderson 2002).

Violence in the household has economic consequences, such as labour absences, job loss and lack of labour force participation (Leone et al. 2004), premature deaths, and costs in health care, as well as the provision of law enforcement and other legal support (Krug et al. 2003). In 2013, nearly US\$800 million were allocated in Mexico to the policy Allocated Expenditure for Women and Gender Equality (CEPAL 2013).

Since the 1990s, international organisations have recognised violence against women as a public health problem (González-Montes 2009).

Since 2007, there has been a General Law on Women's Access to a Life Free of Violence in Mexico, the objective of which 'is to prevent, punish and eradicate violence against women', demonstrating the Mexican Government's commitment to overcoming domestic violence, particularly towards women. However, the data collected by the survey used in this study show that in 2012 about 5.1 per cent of the female population, or almost two million women, and 2.7 per cent of the male population of Mexico suffered some form of domestic violence in the previous 12 months (Engasto 2012).

The present study is based on two hypotheses. The first is that there is a significant effect of domestic violence on individuals' subjective wellbeing. The second is that the greatest impact is on women. The following section describes the data used in the study.

3. Data and Methodology

In Mexico several surveys investigate the issue of domestic violence. The 1998 National Survey on Reproductive Health, the 2006 National Survey on Household Dynamics, which identified greater emotional and economic violence in urban environments and sexual violence in rural areas (González-Montes 2009) and the National Institute of Statistics and Geography (INEGI), which began in 2012, incorporating questions on happiness and satisfaction with life through questionnaires, including the 'BIARE' that mainly addresses issues of Self-Reported Wellbeing (Bienestar Autorreportado, in Spanish). The BIARE sample consists of 10,654 people over 18 years old, representing 70.78 million Mexicans. Despite this survey being relatively recent, some studies have used the data provided by the survey (Hernández García and Tomé González 2016; Temkin 2016).

The present study is based on the BIARE survey. To construct violence variables, questions are asked in relation to: 'during the last 12 months did you suffer from physical aggression by someone with whom you live?'; 'Any other known person?'; or 'Some unknown person?'. Three similar questions are asked about threats. Hence there are six

variables linked to this type of abuse. There are no data on household income, so it was decided to use family spending as a proxy. Regarding happiness, the question was 'On a scale of 0 to 10, how happy would you say you are?' The family variable was generated from the question 'How many people, who are in your family, always will be available to help you in an urgency?' In the case of friends, the question was asked: 'How many people, who are not in your family, always will be available to help you in an urgency?' Marital status was represented by dummy variables *de facto*, separated, divorced, widowed and married; labour participation and educational level were also generated from dummy variables.

Table 1 shows the descriptive statistics for the survey. It is observed that men and women are similar in terms of number of friends, family and marital status. There is a higher rate of employed men than women and 36 per cent of women report household work as their main activity. There is no significant difference between the sexes in terms of level of education. Table 2 shows that women suffer domestic violence to a greater degree than men. Table 3 supports the first hypothesis, showing a significant difference in subjective wellbeing between men who suffer violence and those who do not; the results are similar for women. Comparing men and women, it is shown that women's subjective wellbeing is negatively affected to a greater degree by violence than men's.

The model uses OLS to explain the happiness of individuals. Although the possibility of using other models (e.g., an ordered probit) was considered, previous studies have shown that the results of the OLS are acceptable (Ferrer-i-Carbonell and Frijters 2004).

Happiness economics studies have addressed many variables to compare the level of satisfaction reported by individuals. This study aims to investigate the effect of domestic violence and to compare it, in terms of magnitude, with other elements causing unhappiness, such as unemployment and disability. In this study, it is expected that domestic violence affects women more than men. As control variables we use age, expenditure decile and size of locality, number of friends and family

Table 1 Descriptive Statistics of the Variables Used in the Study, Mexico 2012

<i>Variable</i>	<i>Description of the variable</i>	<i>Average of men</i>	<i>Average of women</i>
Family	Quantity of relatives who gives support	3.71	3.72
Friends	Quantity of friends who gives support	2.45	2.33
Civil union	People related by civil union	0.18	0.16
Separated	People separated from his couple	0.05	0.08
Divorced	Divorced	0.02	0.03
Widow	Widows	0.02	0.06
Married	Married people	0.48	0.45
Employee	People with a job	0.86	0.56
Unemployed	People without a job	0.04	0.01
Household	Household work	0.01	0.36
Student	Students	0.04	0.04
Disabled	People disabled	0.01	0.00
Indigenous	People who speak indigenous language	0.17	0.15
Without schooling	Without schooling	0.04	0.06
Primary school	Primary school ended	0.25	0.28
Secondary school	Secondary school ended	0.29	0.26
High school	High school ended	0.23	0.25
Bachelor's degree	Bachelor's degree ended	0.18	0.14
Postgraduate studies	Bachelor's degree ended	0.02	0.01
Age	Age	39.51	39.52
Size of city (1)	City with 100k of habitants or more	0.49	0.48
Size of city (2)	City with 15k, and less of 100k of habitants	0.24	0.25
Size of city (3)	City with 2.5k, and less of 15k of habitants	0.13	0.13
Size of city (4)	Village with 2.5k of habitants or less	0.14	0.14

Source: Engasto (2012).

giving support, indigenous language, marital and labour status and educational level.

4. Results

Table 4 shows that, in 2012, aggression, particularly within the household, reduced

women's happiness, suggesting that while happiness economics studies have focused on issues of income, poverty, employment status and macro-economic factors, they should also consider what takes place within households, since it is there, in everyday life, where the variables related to happiness have a greater impact.

Table 2 Percentages of People Suffering Aggression and Threats, Mexico 2012

		<i>Men</i>	<i>Women</i>	<i>Average</i>
Aggression inside the household	No	97.29	94.70	95.84
	Yes	2.71	5.30	4.16
Aggression from someone known	No	95.47	96.29	95.93
	Yes	4.53	3.71	4.07
Aggression from someone unknown	No	92.50	96.91	94.97
	Yes	7.50	3.09	5.03
Threats inside the household	No	98.29	97.18	97.67
	Yes	1.71	2.82	2.33
Threats from someone known	No	96.00	96.22	96.13
	Yes	4.00	3.78	3.87
Threats from someone unknown	No	91.56	95.48	93.76
	Yes	8.44	4.52	6.24

Source: Engasto (2012).

Table 3 Average Happiness Reported by Men and Women who are Victims and Not Victims of Violence, Mexico 2012

Sex	Aggressions			Threats		
	Inside the household	Known	Unknown	Inside the household	Known	Unknown
General average no victims	8.45***	8.44***	8.43***	8.43***	8.43***	8.43***
General average victims	7.26***	7.59***	7.85***	7.22***	7.73***	8.02***
Men no victims	8.46***	8.46***	8.48***	8.45***	8.45***	8.48***
Men victims	7.76***	7.99***	7.92***	7.82***	8.07***	8.00***
Women no victims	8.44***	8.42***	8.39***	8.41***	8.40***	8.38**
Women victims	7.02***	7.10***	7.69***	6.86***	7.38***	8.05**

Source: Engasto (2012).

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 4 Determinants of Happiness, Beta Coefficients, Mexico 2012

Variables	Men	Women
(Constant)	8.227***	7.645***
Aggression inside the household	-0.334*	-0.920***
Aggression from someone known	-0.270**	-0.735***
Aggression from someone unknown	-0.244**	-0.401***
Threats inside the household	-0.553**	-0.560***
Threats from someone known	-0.356***	-0.443***
Threats from someone unknown	-0.356***	-0.074
Family	0.115***	0.097***
Friends	0.055***	0.075***
Single	-0.245***	-0.099
Widow	-0.105	-0.285***
Consensual union	0.048	0.063
Unemployed	-0.336**	-0.514***
Retired	0.171	0.026
Household	-0.328	0.067
Student	0.157	0.137
Disability	-1.546***	-1.012**
Kinder	0.000	2.137
Primary	0.077	0.12*
Secondary	-0.003***	0.071
Normal	-1.029	0.112
Technical and primary school	0.435	0.114
Technical and secondary school	-0.055	-0.123
Technical and secondary school	-0.112	0.389**
Normal and high school	0.311	0.243
Bachelor's degree	0.011	0.136
Master's	0.030	0.177
PhD	0.308	-0.012
Decile	0.035	0.057
Age	-0.011	-0.004
Locality	-0.011	0.013
N	4618	5894
R-sq	0.078	0.094

Source: Engasto (2012).

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

In the case of men, it is seen that both aggression at the hands of known people and receiving threats from strangers are associated with a significant reduction in happiness. As regards the goodness of fit of the model, R^2 is relatively low, as has occurred in other studies focused on subjective wellbeing (Ferrer-i-Carbonell and Frijters 2004; Stutzer 2004).

This section describes the results of the estimates of the elements that affect the happiness of individuals. First, it is observed that the presence of friends and family contributes positively to happiness, both for men and women. With regard to marital status, being single or a widow are elements that cause unhappiness for men and women respectively. Having a disability that limits individuals' labour force participation is negative for both sexes, but more so for men, because in Mexican culture men are considered to be the breadwinner. Most of the variables referring to schooling are not significant, except for a positive effect for women who have technical training.

In reference to the main variables of interest, those corresponding to types of aggressions and threats, the hypotheses are confirmed. Violence influences the subjective wellbeing of people. Additionally, five of the six types of violence evaluated have a greater effect on women than on men (Table 4). For women, the variable 'aggression within the home' represented a coefficient of -0.9 . In the case of men, the impact is -0.3 . This shows that women are more sensitive to domestic violence, in addition to suffering it to a greater degree. Assaults by someone known, but not within the home, also represent negative effects on the wellbeing of men and women, with women most impacted. Threats from someone in the family represent a similar effect for both men and women, close to -0.55 , which shows that verbal violence also causes harm to individuals. Threats by members of other households is the fifth significant and negative variable for both sexes. This type of violence suggests that socialisation with others can be paradoxical, being both a source of gratification and happiness, but also potentially exposing individuals to psychological violence. Threats by strangers had a negative impact on the

happiness of men, but not on women, which may be a characteristic of the Mexican context in the early 2000s, in which a war has been waged by the government against groups of drug cartels triggering crimes and extortion in many Mexican cities (Aroche Aguilar 2017).

To complement the work, we estimate some determinants of the probability of suffering violence based on the individual characteristics of the respondents. Table 5 presents the results of the estimates for violence and threats from a binomial probit model.¹ For this model the number 1 represents that the individual suffered violence, and 0 where no violence has been suffered.

It is interesting that the percentage of correctly specified values in this study was between 94 and 98 per cent. Thus, we can observe a high level of prediction of the variables that affect domestic violence.

The results show that having social support reduces the chances of a woman suffering aggression from the person with whom she lives. In the case of men, the result was not significant. With regard to threats, it is observed that a greater number of friends does not imply a lower probability of receiving threats from someone inside the home. This may be because threats are a type of aggression more difficult to observe by others. As previously noted, violence does not end on its own, normally an external agent is required to stop it. In this way, when a woman has more friends who are willing to support her in difficult situations, the opportunity to receive help and guidance is more likely.

The family variable had a more significant impact than friends. For men, having more family members supporting them reduces the possibility that they are victims of aggression with the home (no significance for friends). In the case of women, the presence of more family members is a more important element in reducing the probability of aggression with the home than the presence of friends, consistent with the importance of family over friendships in Mexican culture (Charles-Leija, Aboites and Llamas 2017). Thus, family support is a key component in reducing the likelihood of a woman being the victim of

Table 5 Determinants of Physical Aggression and Threats by People Living in the Same Household, Distinguished between Men and Women, Mexico 2012

Variables	Aggressions		Threats	
	Men	Women	Men	Women
Family	-0.058**	-0.047***	-0.016	-0.055***
Friends	0.016	-0.038**	-0.018	-0.028
Widow	0.000	0.0134	0.000	-0.070
Divorced	0.120	0.249	0.221	0.184
Consensual union	0.453***	0.341***	0.303**	0.344***
Married	0.169	0.182**	0.0132	0.131
Away from his/her partner	0.573***	0.425***	0.621***	0.455***
Employee	-0.325	0.287	-0.386	-0.097
Unemployed	-0.335	0.278	-0.264	0.215
Retired	-0.180	0.086	-0.286	-0.193
Household	0.489	0.104	0.469	-0.252
Student	-0.282	0.294	-0.0941	0.000
Disability	0.000	0.400	0.000	0.000
Other situation in job	0.000	0.000	0.000	0.000
Ln expenses in household	-0.021	0.006	0.096	-0.059
Indigenous	0.153	0.261***	0.240**	0.178*
No education	0.143	0.566*	-0.401	-0.011
Primary school	0.216	0.323	0.166	-0.185
Secondary school	0.040	0.401	0.024	0.099
High school and technician	0.159	0.219	-0.0167	-0.022
Graduate	0.025	-0.021	0.053	-0.173
Postgraduate	0.000	0.000	0.000	0.000
Age	0.003	0.011	0.005	0.022
Age squared	-0.000	-0.000	-0.000	-0.000
Size of city	-0.001	-0.023	0.031	-0.028
Constant	-1.523	-2.257***	-3.124***	-1.400*
N	4,530	5,931	4,529	5,877
Pseudo R-sq	0.041	0.05	0.043	0.045
Correctly classified	0.972	0.947	0.982	0.972

Source: Engasto (2012)

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

domestic violence. For both men and women, having more supportive relatives does not imply a significant change in the propensity to receive threats from someone within the home.

The results show that an individual's employment situation does not have a significant effect on the incidence of violence, suggesting that despite having income a woman can be a victim of violence within the family. Some studies have identified that women entering the labour market can be a trigger for violence, since this represents a 'challenge' to male authority (Krug et al. 2003).

Regarding indigenous language, the result reinforces the perception that indigenous women are a highly vulnerable group. In the case of men, the result shows that being

indigenous does not increase their possibility of suffering domestic violence. With regard to threats, their likelihood increases for those speaking an indigenous language, for both men and women.

The regressions show that for men, level of education has no impact on their likelihood of suffering domestic violence. For women, on the other hand, the effect is significant for those with a technical and bachelor's degree. Both cases have a negative impact on their likelihood of being victims of violence. With regard to threats, it can be seen that men who only completed primary school are more likely to be threatened, as well as those who have technical studies with primary or basic primary education. Contrary to the above, women who have

completed primary education are less likely to receive threats.

5. Discussion and Conclusions

Our results support the study's hypotheses. Violence influences subjective wellbeing, and the effect is greater for women. Also, it was found that social and family networks have negative effects on the likelihood of physical assaults or threats. Speaking an indigenous language has a greater likelihood of being subject to violence, however, surprisingly, women who participate in the labour force are not less likely to suffer domestic violence in Mexico.

The number of friends reduces the possibility of suffering physical aggression for women, but not for men. Social networks are important in a range of ways, for example, according to data from the National Survey on Occupation and Employment, more than 50 per cent of respondents found their jobs via personal contacts (Charles-Leija, Torres and Castro 2018). Social support has the potential to reduce domestic violence, with prior studies establishing that greater social support decreases the likelihood of women suffering domestic violence and can diminish the risk of suicide in abused women (Kaslow et al. 1998; Mitchell and Hodson, 1983). In some cases female victims of violence respond by isolating themselves (Mitchell and Hodson 1983), perhaps due to shame or fear of reprisals (Coker et al. 2002), or perhaps the aggressor has attempted to isolate the woman to maintain greater control over her. Also, social capital allows individuals to have greater access to information (Stigler 1961), which can be useful, for example, in finding a job, seeking help, and so on.

The article identifies that family support is a strong deterrent of violence against individuals. First, family support significantly reduces the likelihood of domestic violence for both men and women. It also reduces the possibility of suffering violence at the hands of people known outside the home. It should be noted that the variable 'Family' indicates the number of relatives willing to support, not those who

necessarily live in the same household as the respondent. However, the presence of such individuals suggests closer relationships, as well as greater emotional stability and support. The effect of the family to limit aggression on the part of known persons is greater in the case of the women than men. As for violence from strangers, the family network again contributes significantly to the safety of women. Family is a key element for Mexicans, offering support, protection, shelter, encouragement and food or income in times of unemployment.

As an example of the importance of social norms in Mexico, integration of the indigenous population has had a positive impact on the likelihood of suffering violence, most likely because of a culture of discrimination in Mexico that has existed since colonial times. Indigenous people usually have low incomes and this contributes to their vulnerability. In the indigenous context, violence is seen as acceptable, and along with poverty this is one of the biggest problems faced by women living in rural areas. Family and friends of indigenous women suffering domestic violence are less likely to intervene since within this culture the head of the household is considered as having the authority to use violence as a form of discipline. Even local authorities may view domestic violence in this way (González-Montes 2009).

In this research we identify a family paradox, in which the household can be a source of happiness, but also pain. It has been documented that a good family relationship is a key component of people's happiness (Becchetti and Pelloni 2008; Winkelmann 2005). For many people the home is the place where they can receive emotional support and share life's best moments. However, in personal relationships there may also be conflict, in the worst cases, violence. Becker (1974) proposed that the family produces goods such as love, sex and affection, but for some families the products are punishment, mistreatment and aggression. The issue is complex and different recommendations can be made in terms of public policy and individual behaviour. Capturing information that relates the incidence of violence to the

personal wellbeing of individuals is a first step. In the 2014 Self-Reported Wellbeing Survey, the number of questions linked to violence and threats received was reduced from six to one, meaning that less useful data is now collected.

The findings of this study suggest that one policy implication is the promotion of social networks for individuals, as it was observed that among the main deterrents of violence were family and friendship networks. Undoubtedly the issue deserves more attention because the percentage of women suffering domestic violence is high and the impact on their wellbeing very significant. Similarly, working to reduce wage inequality by gender will provide women with better opportunities to gain economic independence. Other public policies that may reduce violence against women include simplifying divorce proceedings (Stevenson and Wolfers 2006) and creating work groups formed exclusively by women, where their members can provide protection and secure social support (Krug et al. 2003).

Reduction in violence against women would have repercussions at an economic level, but even more so in terms of happiness. Considering that the lower the level of schooling, the more likely a woman will be a victim of violence (Moral de la Rubia and Ramos Basurto 2016), a relevant approach for governments would be to provide educational programs that allow women better defence mechanisms against domestic violence. Likewise, it is fundamental to work in the promotion of a culture of values that promotes equality between men and women.

Endnotes

1. See Greene (2003) for more information on the binomial probit model.

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